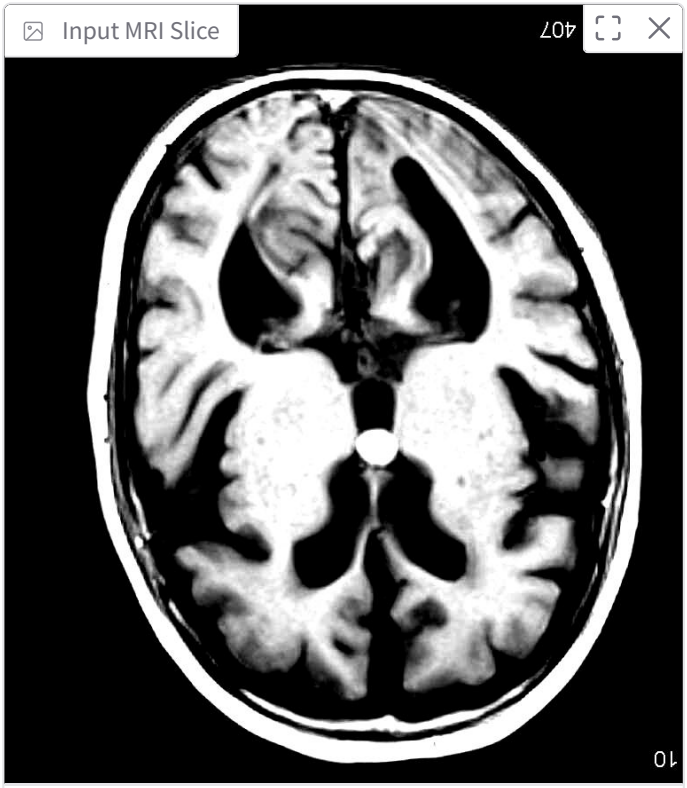
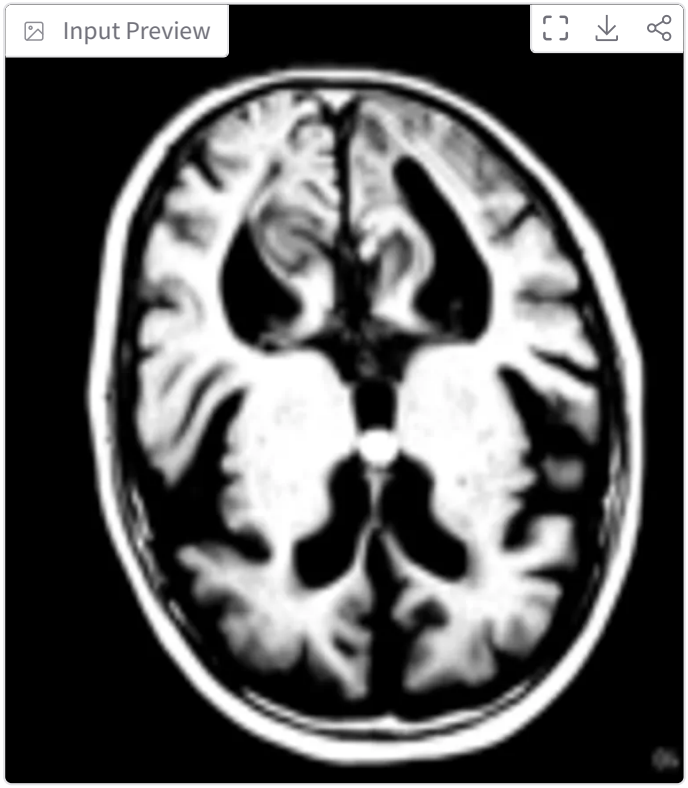
# Neural Anisotropic Diffusion Demo

Upload a brain MRI slice, run the learned PDE denoiser, and inspect the diffusion trace.

Input MRI Slice




Input Preview



Denoised Output

Checkpoint Path

checkpoints/best_model.pth

Neighbor Mode

Iterations 10

1 32

Lambda 0.1

0.005 0.125

Use Multi-Scale Diffusion

Dropout for Uncertainty 0.1

Device: `mps`

Checkpoint

0 0.5

Model Input Size 128

64 256

Trace Capture Stride 1

1 4

Uncertainty Samples 8

2 16

- [ ] Use Residual Refinement
- [ ] Use MiniUNet Guidance

Checkpoint:

`/Users/tushar/Documents/Repositories/neura anisotropic-diffusion/checkpoints/best_model.pth`

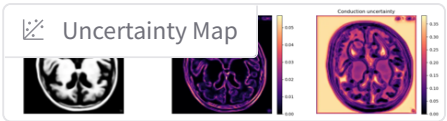
Neighbor mode: `4` | Iterations: `10` | Lambda: `0.1`

Multi-scale: `False` | Dropout: `0.1` | Uncertainty samples: `8`

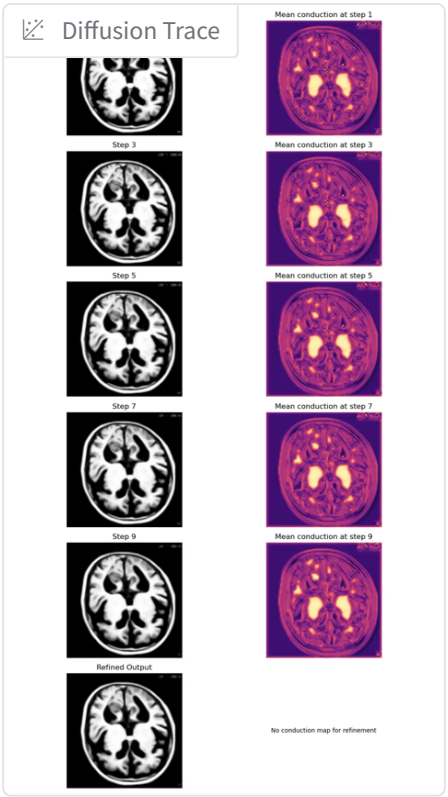
Mean conduction: `0.3951` | Mean update magnitude `0.0325` | Output uncertainty: `0.0065`

**Run Diffusion**

Uncertainty Map



Diffusion Trace




Conduction Summary

Use via API · Built with Gradio · Settings